ECE 198 speaking script

Slide 1 - Exercise tracking for CCCARE

0:00 to 0:35 - Alvern

One instructor can be responsible for up to thirty people in a CCCARE graduate exercise session. While that instructor helps one person, someone else may lose count or move faster than their plan allows. We're PBVR, and our project focuses on that gap in supervision. We want to give instructors a clearer record of each person's set. We've developed two concepts to explore: a wearable motion tracker and a fixed distance station.

Slide 2 - A clearer record of each set

0:35 to 1:12 - Brandon

The problem is that instructors can't easily check everyone's repetitions and pace. Our broader goal covers selected lower- and upper-body exercises. We'd confirm seated knee extensions with an instructor before comparing both concepts on that movement. Participants would see a count, while staff would see cycles needing review. Motion rules would come from selected development data, with settings chosen for the exercise plan. Staff would still judge technique. The first concept collects that information through a wearable sensor.

Slide 3 - Concept 1: Wearable motion tracker

1:12 to 2:05 - Ronnie

For the wearable, we'd collect controlled test movements during development to set repetition and timing rules. These would run on the Arduino. For the first prototype, an accelerometer would be strapped to the shin. A short still reading would establish a resting baseline. The Arduino would use tilt changes to count each outward movement and return as one cycle. The display would show the count and flag unusual cycles.

Strap slip could still distort readings, and one sensor can't establish exact joint angles. We'd secure the electronics at the waist and keep the cable close to the leg. The alternative moves the sensor off the body.

Slide 4 - Concept 2: Fixed distance station

2:05 to 3:00 - Vinay

At that station, an ultrasonic sensor sends sound pulses toward the moving limb. The returning echo gives a distance reading. The Arduino would track distance over time to count movement cycles and check their pace. With a fixed mount and a marked position, the participant would press start for a resting-baseline check. Staff would select the exercise profile, with movement rules developed through earlier testing.

The leg would still need to stay in the sensing area. Clothing or another limb could interrupt the echo, and distance alone can't reveal posture. Testing would need to show whether those readings are reliable. We can compare both designs using the same Arduino kit.

Slide 5 - Both concepts use the same base kit

3:00 to 3:40 - Alvern

That kit supplies the board, display and pushbuttons, with breadboard wiring. The wearable needs the accelerometer listed here. The station needs the ultrasonic distance sensor. Both are available to borrow, but quantities are limited. We'd test them one at a time using our single board. We'd source a strap or mount, an enclosure and power connections. The wearable also needs portable power. Our fifty-dollar budget is a target, with extra parts still to price. With hardware largely shared, the comparison comes down to setup and flexibility.

Slide 6 - The wearable is our starting preference

3:40 to 4:20 - Brandon

The wearable could move from the shin to the forearm for a movement such as an arm curl. That would need a separate exercise profile and testing. The station would need positioning for each movement. Both would use preset rules and a resting baseline, with limits suited to the person's plan. We favour the wearable's placement flexibility, but we'd need to prove that switching exercises is practical. Our comparison would focus on setup time and counting accuracy.

Slide 7 - What we need to prove next

4:20 to 4:38 - Ronnie

We'd compare each count against a manual count on movements we hadn't used to set the rules. We'd include partial movements and pauses, then check missed or extra counts. We'd also measure setup time and check display readability.

4:38 to 4:55 - Vinay

Our first milestone is one working prototype for one exercise. From there, we'd test additional movements and placements. The aim is a count participants can follow and a short summary that helps staff decide what needs attention.

4:55 to 4:57 - Slide 8: Display sources. No narration.

4:57 to 5:00 - Slide 9: Display AI disclosure. No narration. End recording at 5:00.